

The ATA Frequency and Time System

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1. Requirements, Goals, and General Design of the ATA Frequency and Time System

Figure 1 is a simplified block diagram of the ATA Frequency and Time System (FTS). In this first section we discuss the requirements and goals of the FTS, and the basic design choices that were made to satisfy them. The next section discusses the individual components in more detail. The final section elaborates the interface to all other ATA systems. *Table 1* provides a short summary of the requirements and goals.

1.1 Frequency and Time

The maximum phase rate for the ATA is 1233 Hz and the minimum delay step (assumed to be 1 ns) interval is 0.91 ms (D'Addario 2001). Therefore to correct the sampled voltage stream from an antenna to 10^{-1} radian in phase and 0.5 delay steps requires that the device making the corrections can relate the position of the samples to time with an accuracy of $(10^{-1}/(2\pi 1233)) = 13\ \mu\text{s}$ for phase and $(0.91*0.5) = 0.46\ \text{ms}$ for delay. Wideband interference nulling may require updates at approximately ten times this rate in the worst case scenario (Harp, 2004). The circuits that actually perform the operations required on the data are small and fast, so it seems reasonable to assign a peak error *requirement* for the time distributed by the FTS of $10\ \mu\text{s}$ with respect to UTC (with a goal of $1\ \mu\text{s}$ to support nulling).

For pulsar timing the most accurate possible local version of TAI that we can afford is desired. At present, with SA off, the state of the art in time transfer using GPS is of order 20 ns (Clark 1991), so we adopt as a *goal* the ability to relate the internal “ATA time” to TAI (via GPS) to be $\leq 20\ \text{ns rms}$. As noted in the previous paragraph, distributing timing signals at this accuracy is not required for proper operation of the array, but the design presented here is expected to provide the measurements necessary to relate ATA time to TAI in post processing (this step will include perforce retrospective corrections

between GPS time and TAI of order 10 ns). It will be the responsibility of the user to apply the data provided by the ATA if their accuracy needs are better than 10 μ s.

The interstellar medium limits the coherence of extraterrestrial beacons; at worst this limits the coherence length of such signals to ~ 10 s, while at best the coherence time is $> 10,000$ s. The ionosphere also limits coherence for signals at ~ 1 GHz to of order 100 to 1000 s; at higher frequencies the time standard used is the limiting factor. Experience has shown that only a standard with the stability of an H maser (or better) is suitable for VLBI above 1 GHz (Thompson 1986); a similar conclusion can be derived for SETI detections with coherence times ≥ 100 s.

SETI detectors will estimate CW signal frequencies to an accuracy of ~ 10 mHz, this implies the desirability of a standard with frequency known to $\sim 10^{-11}$ or better.

The considerations above mandate that the FTS use an H maser for the primary frequency standard and GPS for time transfer. The design of the ATA is to maintain an ATA time based on the H maser (because it is much more stable than GPS on the timescales of interest to observers) and to monitor the frequency and time offset of ATA time against GPS time provided by a suitable average of the 1PPS outputs of a GPS receiver.

Distribution of ATA time occurs *via* three routes: 1) time accurate to 1 ms is distributed to all computers over the LAN, 2) an accurate (see next paragraph) ATA 1PPS is distributed over fiber to devices requiring precise synchronization to ATA seconds, and 3) a ~ 400 kHz “fast synch” signal is provided over fiber to devices requiring a finer division of ATA time. The “customers” for the more accurate timing signals are the 8 Digitizer racks, the 8 IF Processor (IFP) racks, the 4 to 8 “F card” racks of the Correlator, the 8 Phased Array Network Data Interface (PANDI) cards, other devices (TBD) that need to use the BET/BER data transfer links, and a few miscellaneous devices. The antenna control computers do not need time with more accuracy than that provided over the LAN (1 ms).

Although the frequency of the H maser is stable to 1 part in $\sim 10^{14}$ on time scales > 10 s, its average frequency can only be set to a fractional accuracy of about 10^{-12} , and may be significantly worse if the maser has not been maintained recently (Bower 1997); thus the drift of the ATA 1PPS, driven by the maser, against TAI will generally be ~ 100 ns per day, and to maintain the offset below the goal of 1 μ s may require resetting the ATA 1PPS at roughly weekly intervals. If operational experience shows that we need to resynchronize the ATA 1PPS more often than we prefer, we could decide to let the offset between the ATA 1PPS grow to greater than 1 μ s – in that case tasks that require greater real-time accuracy would need to use the Control System data acquired from the interval meter, probably by sending a request to a time-server task. To the ultimate accuracy the ATA will be able to provide, the offset should be constant over hours, so frequent updates of the estimated offset would not be required in any scheme.

1.2 Local Oscillators

The FTS is also required to provide the local oscillator signals, locked to the maser standard, necessary to support the ATA frequency conversion scheme (*Figure 5*). The four LO1 sources must be tunable over the range 16.3 to 26.6 GHz. LO2 and LO3 are fixed oscillators at frequencies of ~ 14 and ~ 1.5 GHz; for reasons discussed in the next section the precise frequencies selected are 14.0616 GHz exactly and 1562.4 MHz exactly. Distribution of these signals to the RF/IF cards and to the Digitizers is the responsibility of those systems.

It is difficult to provide a rationale to support a particular phase noise specification for these local oscillators. Lacking that we set as our requirement “good standard practice.” Experience at many observatories has shown that the specifications of good quality off-the-shelf oscillators are sufficient for cm-wave arrays. The requirement for “good practice” will also extend to the distribution of these signals and the reference from which they are derived.

More problematic are the presence of discreet spurs in the LO signals. If these spurs are in a frequency interval that can enter the IF system, they will produce two undesirable effects. First, they will be detectable by the coherent SETI detectors at levels above -10 dB with respect to the system noise in 1 Hz (with a typical $\sim 10^2$ s integration). However, although detectable internal spurs are not desirable for SETI, as long as they are not too common the observing protocols used by the SETI system will reject them as part of the much greater (we presume) numbers of external RFI signals with very little extra adverse effect. Second, spurs may produce errors in the cross-correlations that prevent averaging down to the noise on long integrations. Such spurs can also cause closure errors. The errors in the cross correlations will have the worst effects at the narrowest channel bandwidths. The narrowest channels used for astronomical observations will be of order a few km s^{-1} , corresponding to a ~ 10 kHz channel bandwidth at 1 GHz. In a 10^5 s integration the noise should be averaged down in such a channel by 45 dB, so we would like the spurs to be at least 55 dB below the noise in 10 kHz, or equivalently 15 dB below the noise in 1 Hz. For continuum imaging, the presence of spurs could limit the dynamic range achievable. In the lower part of the ATA RF range, continuum observations will be confusion limited after ~ 100 s of integration; for a single 100 kHz channel such an integration will beat down the noise by 35 dB, so if we wish the effect of the spur to be at least 10 dB below the noise in any cross-correlation product, then we need to keep the spurs below -45 dB with respect to the noise in 100 kHz, or equivalently $+5$ dB with respect to the noise in 1 Hz. Longer integrations are likely to be used for continuum observations in the upper part of the ATA RF range, but at these frequencies a large dynamic range will not in general be needed as the images are likely to be noise limited. In addition, for integrations longer than 100 s, “fringe washing” will begin to reduce the integrated cross power observed for a spur by a factor that goes as the integration time, while the noise will be reduced by a factor that goes as square root, with the consequence that the noise will dominate the errors introduced by the spurs to an ever increasing extent as the integration time increases. In summary, our goal should be to reduce LO

spurs (at offsets which can enter the signal path) to 15 dB below the system noise in 1 Hz (at the relevant point in the signal path).

1.3 Clocks and Synchronization Signals

Finally, the FTS must provide a clock at $156.25 \text{ MHz} \pm 0.01\%$ both to act as the ADC clock in the Digitizer and the (serial rate / 20) clock required by all devices using the high-speed 3.125 GHz serial fiber data links (used to connect the Digitizers to the IFPs and the IFPs to the Correlator and PANDIs). A clock at (2/3) this rate is also needed by the IFP, the Correlator, and (probably) the PANDIs. All of these locations also need to receive the “fast synch” and ATA 1PPS signals. For reasons discussed in the next section the precise frequencies selected are 156.24 MHz exactly, 104.16 MHz exactly, 406.875 kHz exactly, and 1 Hz exactly (all based on the H maser frequency standard). For proper operation of the serial links the jitter on the 156.24 MHz signal must be < 40 ps peak to peak. The jitter and rise-time on the “fast synch” must be $\ll 6$ ns, the period of the ADC clock. The jitter and rise-time on the ATA 1PPS must be $\ll 10 \mu\text{s}$, the period of the “fast synch”. To provide maximum immunity to noise and isolation between modules, optical fiber was selected as the transmission medium for the two synchronization signals. The stringency of the clock jitter specification, however, led to the adoption of (highly-shielded) coaxial cable using components selected for high isolation. The even-order distortion on these clocks must be low enough that conversion to square waves will result in a duty cycle of $50 \pm 1\%$.

A new requirement, as of revision 3 of this document, is to provide a clock signal of order a few hundred hertz for use by a phase switch. This switch will generate 16 orthogonal Walsh functions that will switch the LO2 phases by π radians and reverse the sign of the corresponding ADC outputs; the result will be to cancel out any signals that leak into the IF between the second mixer and the ADC input. A frequency of 625 Hz was selected (fast sync frequency 651); the ATA 1PPS results from a further division by 625. The full specification of this additional signal is not yet as firm as for all the other signals produced by the FTS.

ATA Frequency and Time Distribution Simplified Functional Block Diagram

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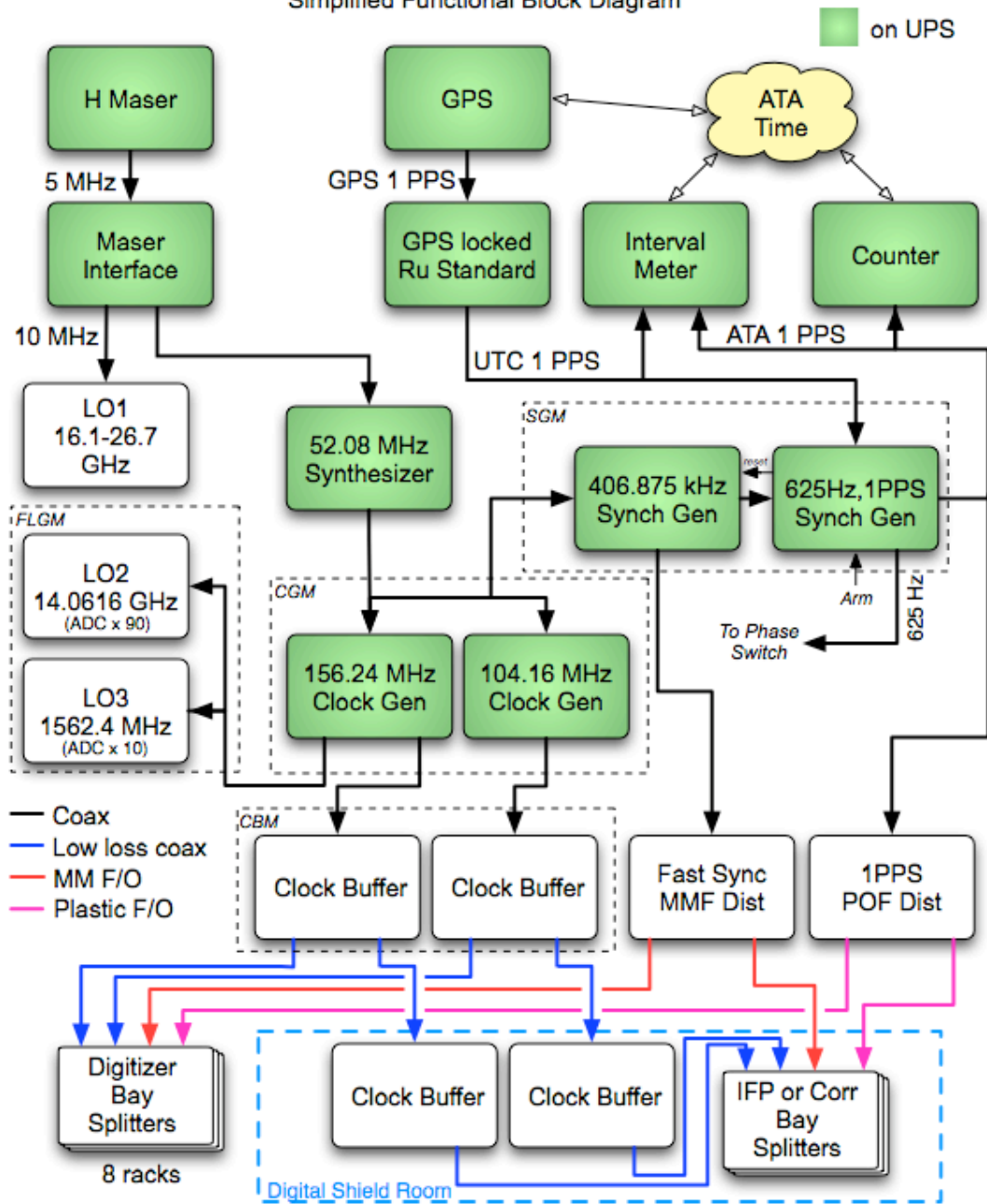


Figure 1. FTS Block Diagram

	Req	Goal	Unit	Comment
ATA Time Error				with respect to UTC
as distributed by NTP	1		ms	peak
as distributed by ATA 1 PPS	10	1	μs	peak
retrospectively available		20	ns	rms
ATA Frequency Error				with respect to TAI
fractional setting error	10	2	10 ⁻¹³	
fractional stability				
10 s interval	-12	-13.6		log ₁₀ of Allan deviation
100 s interval	-13	-14.3		log ₁₀ of Allan deviation
1000 s interval	-14	-14.7		log ₁₀ of Allan deviation
LO1 (four independent versions)				
lowest frequency	16.3		GHz	locked to master standard
highest frequency	26.6		GHz	locked to master standard
resolution	50	0.001	MHz	
output power	5		dBm	
output return loss (match)	15	20	dB	50 Ω
phase noise	*			
non-harmonic spurious	*	-131	dBc	at offsets > 500 MHz
LO2				
frequency	14.0616		GHz	90 x Digitizer Clock
output power	5		dBm	
output return loss (match)	20		dB	50 Ω
phase noise	*			
non-harmonic spurious	*	-115	dBc	1.36 GHz < offset < 1.76 GHz except multiples of
LO3				
frequency	1562.4		MHz	10 x Digitizer Clock
output power	5		dBm	
output return loss (match)	20		dB	50 Ω
phase noise	*			
non-harmonic spurious	*	-115	dBc	50 kHz < offset < 50 MHz
Digitizer and Serial Link Clock				
frequency	156.24		MHz	sine wave (on coax)
level	0		dBm	into 50 ohms
jitter	40	30	ps	peak-to-peak
duty cycle		50	%	± 1%
Decimated Rate Clock				
frequency	104.16		MHz	sine wave (on coax)
level	0		dBm	into 50 ohms
jitter	40	30	ps	peak-to-peak
duty cycle		50	%	± 1%
Fast Synch				
frequency	406.875		kHz	square wave (on MMF fiber)
level	-29	-24	dBm	850nm optical "ON"
jitter	2	1	ns	peak-to-peak
duty cycle		50	%	± 1%
Phase Switch Clock				not firm yet
frequency	625		Hz	pulsed (on coax)
level				TTL compatible
jitter	1	0.01	μs	peak-to-peak, positive edge
pulse length		10	μs	not critical
ATA 1 PPS (based on H maser)				
frequency	1		Hz	pulsed (on plastic fiber)
level	-20	-15	dBm	650 nm optical "ON"
jitter	1	0.01	μs	peak-to-peak, positive edge
pulse length		10	μs	not critical
UTC 1 PPS (based on GPS)				
frequency	1		Hz	pulsed (on coax)
jitter	1	0.01	μs	peak-to-peak, positive edge
pulse length		10	μs	not critical
level		2.5	V	into 50 Ω
* best commercial practice				

Table 1. ATA FTS Requirements Summary

2. Description of ATA Frequency and Time Distribution System

The FTS is described in the accompanying functional block diagram (*Figure 1*) and in the discussion of the modules in that diagram below. The diagram does not show most of the interfaces to the ATA Control System to reduce clutter; what are shown are the inputs used by the ATA Control System to maintain the abstraction “ATA Time”. The Control System is responsible for distributing time to the various control tasks, including those that control hardware that uses the two synchronization signals that the FTS provides. The distribution of the LO signals that the FTS generates is considered to be a function of the RF/IF sub-system; similarly the distribution of the digital clocks and synchronization signals provided to various racks is a function of the relevant system implemented in those racks; formally in fact, the interface between these signals and the FTS is at the end of the cables that connect to the various “customers” of the signals. The dotted boxes indicate a plausible allocation of the indicated functions into physical modules.

2.1 H Maser

The existing hydrogen maser frequency standard (“NX-II”) at the Hat Creek Radio Observatory (HCRO) that has been used for the BIMA array will provide the master frequency reference for all frequencies used in the ATA signal path. Although there is no current set of stability measurements (*e.g.* Allan variance plots) for this unit, experience in general suggests that an H maser should be adequate for use in cm-wave VLBI, and experience with the HCRO maser in particular showed that 1) at 86 GHz the loss of correlation due to short term phase instability was 20% which sets an upper bound on the loss due to the H maser itself that should be more than acceptable for the ATA upper frequency of 11 GHz, and 2) the maser fractional frequency setting error was $\sim 2 \times 10^{-13}$ (Bower *et al.* 1997). The HCRO maser produces 5 MHz and does not produce 1PPS. It has an internal battery backup power source; however the four batteries currently in use should be replaced by two new batteries with greater capacity. The power supply should be moved further from the physics package as well. The maser needs to be located in a section of the “quiet” shielded enclosure which is not subject to drafts, undue vibration, or large magnetic fields.

Given the age of the existing unit, it would be prudent to plan on replacing it when the necessary financial resources can be found (about 200 K\$).

2.2 GPS

Several GPS units are available at the HCRO, including a modern dual-frequency unit. We will evaluate the stability of these units using the H maser and choose the one with the best stability. The chosen GPS receiver will be controlled and read by the ATA Control System. The Control System will distribute UTC over the internal LAN using Network Time Protocol, with an accuracy of ≤ 1 ms. (Hardware “synch” signals at 406.875 kHz and 1 Hz will provide more precise time to hardware as required, see below.) In order to maintain a continuous monitor of the maser frequency the GPS unit should be on long-hold UPS.

2.3 GPS locked Rubidium Standard

An SRS FS725 Rubidium Frequency Standard has been selected to provide a low cost backup in case the rather elderly H maser fails and also to provide a low-jitter (< 1 ns) estimate of UTC seconds. It locks to the 1PPS from the GPS receiver with a time constant of as long as 18 hours. This unit produces several isolated copies of 1PPS, 5 MHz, and 10 MHz derived from the (disciplined) standard. The 1PPS output will align to within 1 ns to the average GPS 1PPS and is denoted “UTC 1PPS” in the block diagram. Full specifications can be found at:

<http://www.srsys.com/products/FS725.htm>

<http://www.srsys.com/products/PRS10.htm>

The stability of this backup standard is not expected to be sufficient for high-quality VLBI at frequencies above 1 GHz (Thompson *et al.*, 1986) and will not allow coherent SETI detection times beyond about 10 seconds (at 3 GHz). The expected Allan deviation as a function of interval is estimated below and compared to the performance of the Parkes H maser (similar in age to the HCRO maser) and to the performance of GPS itself:

Interval (s)	log ₁₀ of Allan deviation		
	FS725	Parkes	GPS
10	-11.4	-13.6	-8.3
100	-12.0	-14.3	-9.2
1,000	-12.4	-14.7	-10.1
10,000	-12.4		-11.1
100,000	-11.6	-14.0	-11.7

This unit also needs to be on UPS. The Control System will monitor the status of this unit.

2.4 Maser Interface

This module accepts 5 MHz from the H maser, to which is locked a high quality, SC-cut crystal 10 MHz “clean up” oscillator that provides a lower phase-noise reference for the microwave frequency chain. We have two Wenzel type 500-04098 oscillators already on hand that will be well-suited to this purpose. Isolated copies of the 10 MHz will be provided for the 52.08 MHz Synthesizer, for the four microwave synthesizers generating the first Local Oscillators, and for general use. This module needs to be on UPS.

2.5 52.08 MHz Synthesizer

The sample data links are all clocked at 156.24 MHz (nearest frequency neatly divisible by 3 within the required frequency range of $156.25 \text{ MHz} \pm 0.01\%$). The $\sim 100 \text{ MHz}$ of IF bandwidth is $\sim 50\%$ over-sampled by the digitizers running at 156.24 MHz; after the fine delay is applied in the IFP units the sample rate is reduced to $(2/3)$ of 156.24 MHz = 104.16 MHz. 52.08 MHz is the greatest common denominator of these two frequencies and, hence, has been selected as the basic frequency for all ATA timing signals. The 52.08 MHz Synthesizer generates a clean analog signal using the maser-derived 10 MHz as a reference. Jitter in the zero crossing must be $< 40 \text{ ps}$ peak to peak. Since 52.08 MHz is not a multiple of 10 MHz, the phase of the output signal with respect to the zero crossings of the 10 MHz reference will be arbitrary and may change whenever this unit is initialized. Since all the ATA timing is derived from this 52.08 MHz, however, such a phase jump will be invisible to “internal” interferometric applications; it *could* be a problem for VLBI or SETI should it reset frequently – hence we have chosen to place this unit (as others) on a long-hold UPS to minimize the chance that it will need to be reset more than once a year or so. (See also discussion in the section on the Interval Meter below.)

The SETI Institute has several high-quality synthesizers that will no longer be needed. Although these are “overkill” for generating a fixed frequency, we plan on using one to begin with since it will require no engineering and is in effect free. The Control System will command and monitor this synthesizer. Distribution of this signal will be on coaxial cable, since 1) none of the signal paths enter the noisy digital part of the shielded enclosure and 2) maintaining high SNR is vital for signals that will be multiplied up (F/O links have terrible noise figures!).

2.6 156.24 MHz ADC Clock Generator and Distribution

This unit uses an analog frequency tripler and bandpass filter to provide a very low-noise, low-jitter sine-wave at 156.24 MHz. This task will be performed by a high-performance commercial unit such as those available from Wenzel Associates. Three isolated copies of signal are made. One is for monitoring, one is provided as a reference for the LO2 and LO3 generators, and one is distributed to the various “customer” racks, where it is “squared up” into a digital clock (with $50 \pm 1\%$ duty cycle). The distribution path consists of 1) a 1x3 unity-gain active splitter with > 40 dB isolation, 2) two ~ 20 m runs of half-inch, low-loss, high-isolation coaxial cable (e.g. Times LMR-500) to the digitizer rack bay and to another 1x3 splitter within the digital shield room (leaving one spare tap on the first distributor), and 3) a passive 1x8 splitter to provide the signal to each of the 8 racks of each bay over smaller coaxial cables (e.g. Times LMR-240). Reverse isolation from the digital shield room to upstream signal paths in the RF enclosure shall exceed 100 dB. **Appendix A** provides a suitable circuit for conversion of the sine-wave clock to a low-jitter differential square wave clock with random jitter below 2 ps rms.

2.7 104.16 MHz IFP Clock Generator and Distribution

This unit is identical to the 156.24 MHz unit except that it uses a frequency doubler rather than a tripler. Distribution of this clock is also similar except that a copy is not needed for the fixed LO generators. **Appendix A** provides a suitable circuit for conversion of the sine-wave clock to a low-jitter differential square wave clock with random jitter below 2 ps rms.

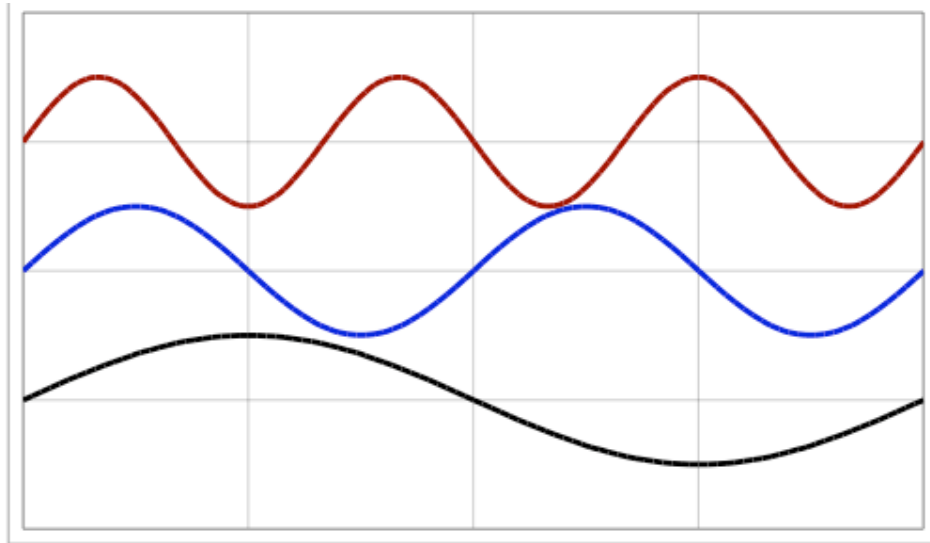


Figure 2. Phase relationship of the 52.08 MHz (lower, black curve), 104.16 MHz (middle, blue curve), and 156.24 MHz (upper, red curve) signals.

2.8 406.875 kHz and 1PPS Synchronization Generators and Distribution

This unit divides down an isolated copy of the 52.08 MHz by 128 to produce a 406.875 kHz “fast synch” square wave (with $50 \pm 1\%$ duty cycle) which is then further divided to produce pulses at 1 Hz rate. The dividers are initialized using the “UTC 1PPS” provided by the GPS-locked Rubidium standard, which provides our best UTC (or TAI) estimate. The initialization to UTC occurs when the Control system “arms” the module (sufficiently prior to the UTC pulse to allow for latency in the network *etc.*). These 1 Hz pulses are denoted “ATA 1PPS” and are locked to the maser rather than GPS. After initialization the first ATA 1PPS pulse will occur 1 second (plus a propagation delay that will be $\ll 1 \mu\text{s}$) later, however the ATA 1PPS will slowly drift versus the UTC 1PPS over days as discussed previously. **Figure 3** illustrates the phase relationship of the fast sync to the two high speed clocks (sections 2.6 and 2.7): the positive edge of the fast synch occurs one half-period of the ADC clock ($\sim 3.2 \text{ ns}$) ahead of the next simultaneous instance of the rising edges of the ADC and decimated clocks. (This relationship will be set by adjusting the relative path lengths to each rack as necessary.) **Figure 4** illustrates the phase relationship of the ATA 1PPS to the fast sync: to minimize the possibility of ambiguity the rising edge of the ATA 1PPS occurs halfway between the rising edges of the fast synch, thus the first rising edge of the fast synch after the rising edge of the ATA 1PPS will occur $1.23 \mu\text{s}$ after the integer ATA second mark.

An intermediate stage of the frequency division of the fast sync down to the ATA 1 PPS is exploited to produce a 625 PPS TTL-compatible signal to drive a phase switching system that has been incorporated into the LO2 Distribution and Digitizer subsystems to reduce possible crosstalk effects.

The fast sync is distributed over multimode fibers using 850 nm LED links intended for 155 Mbaud communications applications; these provide edge rise times of $\sim 2 \text{ ns}$ and pulse distortion $< 800 \text{ ps}$, sufficient to pick out an individual sample at 156.24 MHz. The ATA 1PPS pulses are distributed over 1 mm plastic optical fibers (POF); these simple, inexpensive links will provide a rise time of $\sim 10 \text{ ns}$ with jitter $< 10 \text{ ns}$. The length of these pulses will be $\sim 10 \mu\text{s}$. **Appendix B** provides a suitable circuits for receiving these two optical signals.

This unit needs to be on UPS.

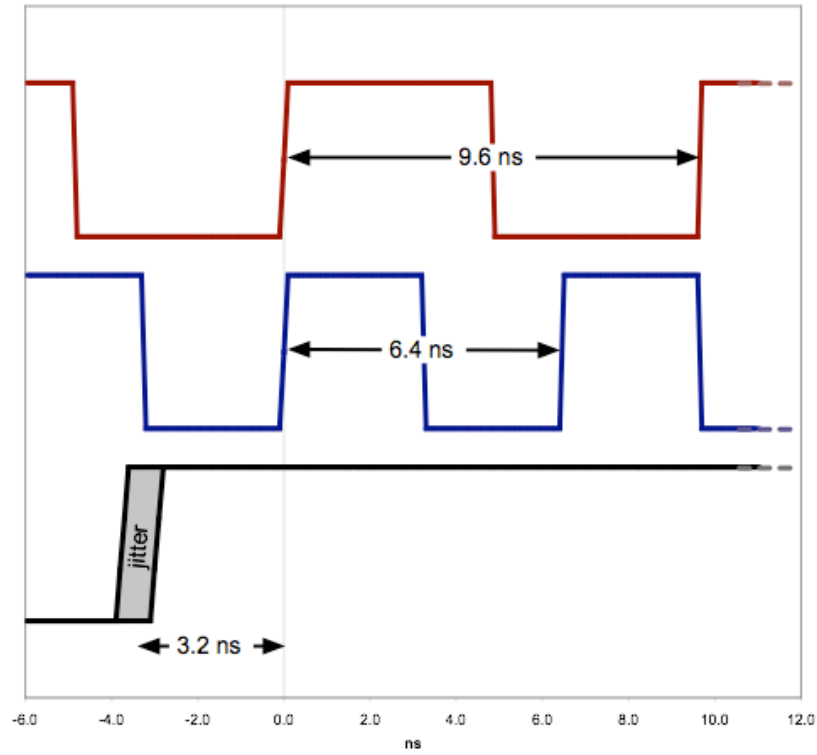


Figure 3. Phase relation of fast sync with respect to ADC (blue) and decimated (red) clocks as seen at the outputs of the recommended receive circuits.

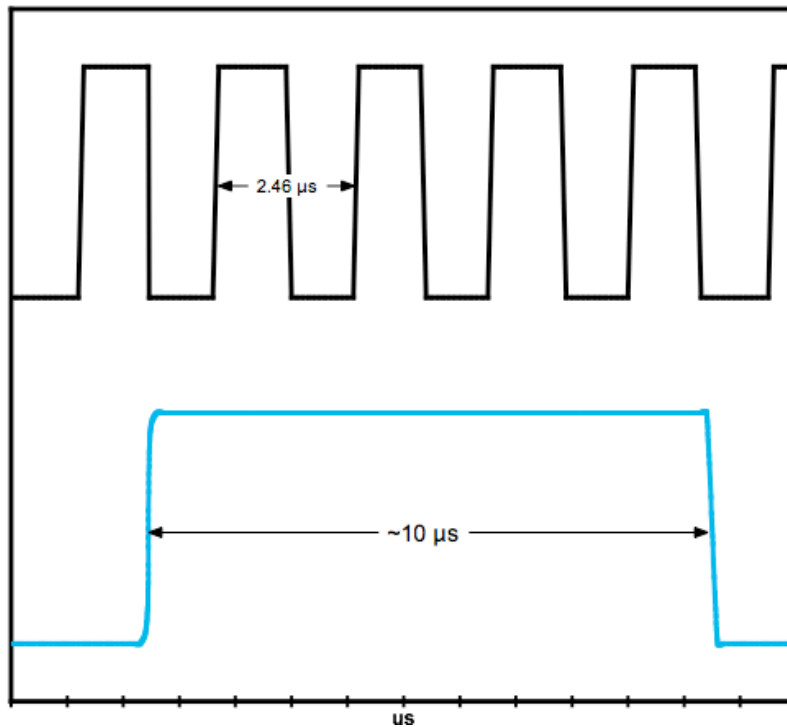


Figure 4. Phase relationship of ATA 1PPS (light blue, bottom) and fast sync (black, top).

2.9 Rack Bay Splitters

These units accept 1 input and provide 8 copies for use in the 8 racks of each bay. The ADC and decimated clocks (on coaxial cable) are split with a passive in-phase power divider. The fast sync (on multimode fiber) is split with a passive fiber “tree” splitter. The ATA 1PPS (on plastic optical fiber) requires the use of a simple active splitter to avoid pushing the link power margins too far.

2.10 Interval Meter

Since the maser relative frequency may deviate from nominal by $\sim 10^{-12}$, the “ATA 1PPS” will gradually drift from the “UTC 1PPS” at the rate of ~ 100 ns/day. Rather than try to correct for this drift, we plan to monitor it with an interval meter that compares the “UTC 1PPS” and “ATA 1PPS” arrival times. Since the relative offset, although small, can be either positive or negative, the intervals measured will either be near zero or near 1 second. The Control System will continuously monitor and archive this data, and make available the (very slowly varying) offset between the “ATA 1PPS” and the “UTC 1PPS” to any devices that require UTC accurate to better than ~ 10 μ s. Users with particularly demanding time requirements should be able to correct the ATA time to estimated TAI in post processing using the archived data with an accuracy set by the long-term transfer ability of the GPS system (including our receiver) – we expect that accuracy to be ≤ 20 ns. The interval meter should have a single shot accuracy well below 1 ns. This unit needs to be on UPS.

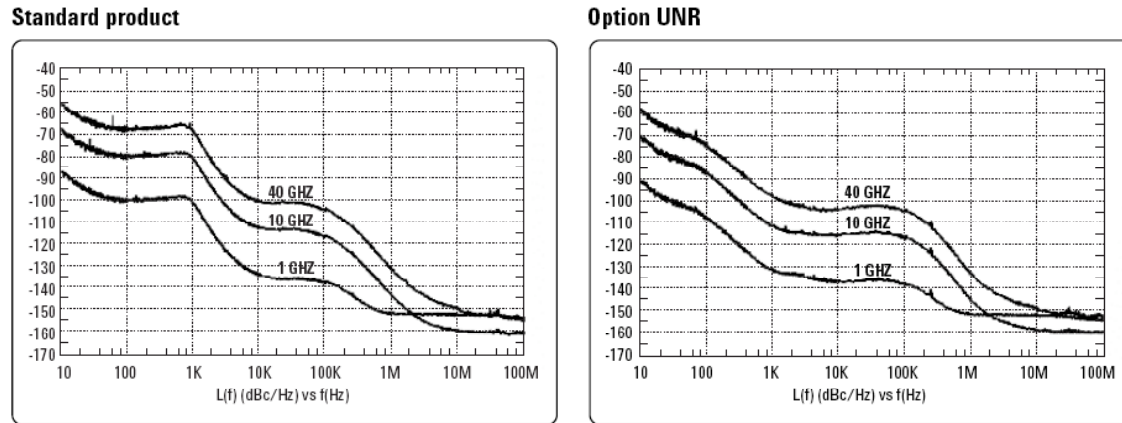
2.11 Counter

This ≥ 10 digit counter keeps track of the total count of “ATA 1PPS” and thus provides a measure of “ATA time”. It is set up, initialized, and monitored by the Control system. This unit needs to be on UPS.

2.12 LO1 16.3 to 26.6 GHz Synthesizer

The first LO must be tunable in steps less than 50 MHz (preferably ≤ 1 kHz) over the range 16.3 to 26.6 GHz or larger (we hope to provide 16.1 to > 26.7 GHz). Four independently tunable versions of LO1 are required. The microwave synthesizers use a copy of the maser 10 MHz as a reference. Whenever these synthesizers are tuned or reset a change in the phase of the output signal with respect to the maser 10 MHz can be expected. This change will have no effect on beam formation or imaging, but is undesirable for phase-referenced VLBI. For this application it will be necessary to forgo tuning LO1 during intervals in which phase must be maintained. The LO distribution

system is included in the RF/IF system rather than the Frequency and Time system. The Control System will command and monitor the LO1 synthesizers. Several vendors make suitable synthesizers, for example the Agilent E8247C whose phase noise curves are as follows:



Meeting the spurious goal for LO1 may be problematic. When tuned to 16.3 GHz the upper edge of the first IF is 0.5 GHz away (see **Figure 5**). As can be seen above, in the range from 10 to 100 MHz offset, the Agilent E8247C, which is fairly typical in this regard, has spurs at about the -150 dBc level. If we estimate that the input power to the first mixer is greater than -130 dBm/Hz (necessary to avoid added noise), the conversion loss is -6 dB, and the LO power out of the IF port is less than -20 dBm (see presentations for ATA Signal Processing CDR, 2003) then the LO leakage will be $+116$ dB with respect to the system noise in 1 Hz. For the spurs to be 15 dB down would then require that they be less than -131 dBc. Although the spurs seen immediately above meet this goal, the actual specification for non-harmonic spurs is only -50 dBc, far short of our goal. We will need to investigate the far-out spurs of the LO1 synthesizer to get a better idea on their real levels – however making measurements down to -130 dBc will be rather difficult. Also, the frequency and level of any spurs may vary with the commanded output frequency, making any kind of exhaustive search impractical. If the performance is found wanting, it should be possible to filter the LO1 signal to remove the undesired spurs. A simple high-pass filter based on a waveguide-beyond-cutoff design should suffice.

2.13 LO2 14.0616 GHz Generator

The second LO is fixed. The frequency is chosen to be the nearest value to the nominal 14.0 GHz that is an exact multiple (90) of 156.24 MHz; this choice ensures that any harmonic of the 156.24 MHz clock (including the 3124.8 MHz serial link clock in the digitizer card) that leaks into the first IF will be translated and aliased to DC in the IF baseband, where it will be removed. To keep any spurious products on this LO 15 dB below the noise in 1 Hz in the offset range that overlaps the first IF (offset frequencies

+1.36 to +1.76 GHz, see *Figure 5*) will require that these spurs be less than ~ -115 dBc. (Here we have estimated that the IF1 signal power at the input to the second mixer will be 20-40 dB below the LO2 drive power to that mixer and that the isolation between the LO port and the input port of this mixer is 20-30 dB.) The spurious level specification of available microwave synthesizers is typically only -60 dBc, although one might expect that any spurs would be unlikely for the frequency offsets of interest to us here. A phase-locked dielectric resonator oscillator (DRO) will have a typical specification of -80 dBc, and the expectation is that the only spurs will be at multiples of the reference, which in our case would be the ADC clock, and therefore harmless as already noted. The most conservative design would be to use a synthesizer or a PLL followed by a narrow ($\sim 3\%$) bandpass filter. The Control System will need to monitor this device.

2.14 LO3 1562.4 MHz Generator

The third LO is also selected to be a multiple of the digitizer clock for the reason noted immediately above. This signal is derived as 10×156.24 MHz using a phase-locked multiplying loop. For special needs the signal could be replaced by a tunable signal from a synthesizer locked to the maser 10 MHz. This LO also requires spurs be less than -115 dBc to meet our goal. The range of frequency offsets for which spurs will be harmful is from 50 kHz (the edge of the DC channel in the correlator) to 50 MHz (the outer edge of the IF band seen by the digitizers). If generated by a phase-locked loop, the only discreet spurs should be either multiples of the reference (156.24 MHz) or multiples of the line frequency (60 Hz), which typically extend to less than a kHz offset. Neither should present a problem. A discussion with one vendor provided the information that the output tone of a coaxial resonator type PLL is clean down to -100 dBc, which is as low as they can measure, except for the reference tone at -90 dBc (typical, specification is -80 dBc). The Control System will need to monitor the lock status.

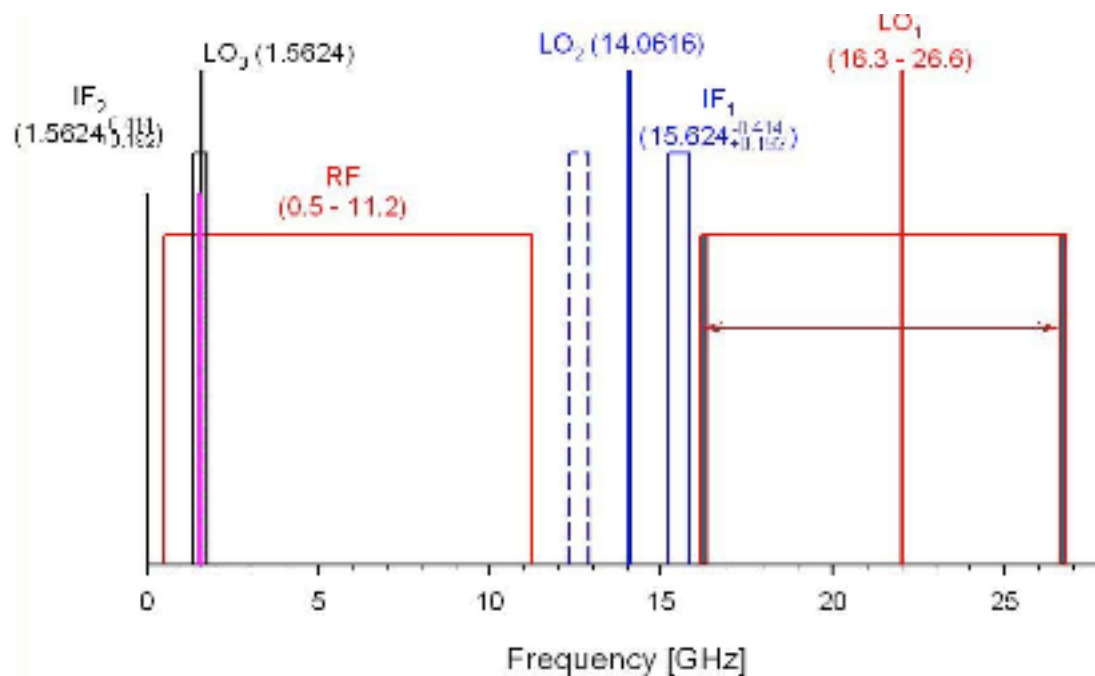


Figure 5. ATA frequency conversion scheme.

3. Interfaces

The FTS interfaces to the Signal Path system, the Control system, and the Site system.

3.1 FTS to Signal Path Interface

The FTS interfaces to the Analog Signal Path subsystem at the RF Converter Rack (RFCR) component of the RF Converter subsystem. The FTS provides the source (but not the distribution) for the LO1, LO2, and LO3 signals:

LO1 is a sinusoid signal tunable from 16.1 to 26.7 GHz with a level of ≥ 5 dBm into 50 ohms with return loss of 15 dB or better on a bulkhead 2.4 mm female connector.

LO2 is a sinusoid signal at 14.0616 GHz with a level of ≥ 5 dBm into 50 ohms with return loss of 20 dB or better on a bulkhead SMA-compatible female connector.

LO3 is a 1562.4 MHz sinusoid with a level of ≥ 5 dBm into 50 ohms with return loss of 20 dB or better on a bulkhead SMA-compatible female connector.

The Phase Switch Clock is a 625 PPS TTL-compatible pulsed signal with pulse characteristics similar to the ATA 1PPS. Full requirements specification is pending. It will be provided on a bulkhead SMA-compatible female connector.

The FTS interfaces to the Digital Signal Path subsystem at the Digitizer and IF Processor racks (eight of each). In addition, the FTS interfaces to those backend devices such as the correlator “F cards” and the PANDI cards (see **Figure 1**) that require synchronization to the synchronous data stream provided by the IFP using the BET/BER protocol. The signals provided are as follows:

The ADC clock is a 156.24 MHz sinusoid with a level of ≥ -2 dBm into 50 ohms with return loss of 20 dB or better on a 0.25 inch coaxial cable with an SMA-compatible male connector. See **Appendix A** for recommended receiver interface.

The decimated rate clock is a 104.16 MHz sinusoid with a level of ≥ -2 dBm into 50 ohms with return loss of 20 dB or better on a 0.25 inch coaxial cable with an SMA-compatible male connector. See **Appendix A** for recommended receiver interface.

The “fast synch” signal is a 406.875 kHz square wave with $50 \pm 1\%$ duty cycle and with jitter and rise-time ≤ 2 ns. It is distributed over 850 nm multimode fiber with ST connectors and received optical power ≥ -29 dBm. See **Appendix B** for recommended receiver interface.

The ATA 1PPS is 1 Hz pulse with rise-time and jitter $\leq 1 \mu\text{s}$ and maintained to be within $\leq 10 \mu\text{s}$ of integral UTC seconds. It is distributed over 1 mm plastic optical

fiber (at 650 nm) with Versatile Link connectors. Optical power received will be ≥ -20 dBm. See **Appendix B** for recommended receiver interface.

All the above signals are driven off the maser standard and are in the same “time domain” in the sense that the relative phases of all the clock edges are fixed.

3.2 FTS to Control System Interface

The Control System monitors and controls several modules within the FTS. Some of these modules are instruments that can be accessed using either TCP/IP, GPIB, or RS232 protocols. Other modules will be controlled using the same protocols but mediated through one or more Control Units, which will probably be commercial devices, *e.g.* the Agilent 34970A. The modules that require supervision by the Control System are:

H Maser alarm status (via Control Unit).

Maser Interface alarm status (via Control Unit).

GPS Receiver status and UTC time information (for NTP server or equivalent) via RS 232 interface. The Control System will use this device to establish the coarse resolution time to disambiguate the ATA 1PPS used by the hardware devices in the signal path.

Interval Meter setup, monitor, and read back via GPIB. The Control System uses this module to keep track of the exact offset of each ATA 1PPS pulse from the UTC 1PPS pulse in order to provide an archive of data that will permit extremely accurate estimation of the frequency of the master frequency standard (maser) and the relation of ATA time to GPS time (and ultimately TAI).

Counter setup and read back via GPIB. This unit keeps track of the count of ATA 1PPS pulses.

52.08 MHz Synthesizer setup and monitor via GPIB. (This may eventually be replaced by a fixed frequency custom unit, in which case only status monitoring via the Control Unit will be needed.)

406.875 kHz and ATA 1PPS Generator must be armed for reset of the ATA 1PPS to the next UTC 1PPS (via Control Unit) when commanded by user (rare event). Also alarm status needs to be monitored via Control Unit.

LO3 monitor lock status via Control Unit.

LO2 monitor lock status via Control Unit.

LO1 setup and monitor errors. Set frequency and amplitude of LO1 as required by observing program. GPIB or 10baseT interface. Select filters (if used in final design) via Control Unit. Four units, one per tuning.

UPS status. Interface TBD.

In addition, it may prove advisable to monitor the rack temperatures and possibly some of the power supply voltages of the custom-built modules using a Control Unit or a similar system multimeter.

3.3 FTS to Site System Interface

The FTS system will require two racks in the quiet shielded enclosure. One of these is the existing H Maser rack, which needs to be located in a low-traffic area and protected from stray magnetic fields and excess vibration. Much of the FTS must be on long-hold UPS (see Figure 1). The LO and clock generators will be located in the LO Source Rack (LOSR) of the RFC subsystem (about 32U excluding the UPS). Total AC power draw should be < 2 kW at 110V 60 Hz. The long-hold UPS will need to supply at most 1 KVA. The components of the FTS will be most stable if the environmental temperature can be regulated to vary as little as feasible, preferably less than 1 degree centigrade peak to peak over one day.

Control System signals (probably a single duplex optical fiber) will need to reach the FTS from the digital shielded enclosure. Two half-inch coaxial cables and 6 optical fibers will also need to pass from the quiet side of the shielded enclosure to the digital side.

The FTS will require a GPS RF signal from the site GPS antenna.

References

Bower, Geoffrey C., Backer, D. C., Gibson, J., Plambeck, R., 1997, "April 1997 3mm VLBI at Hat Creek," BIMA Memo #52.

Clark, Tom, 2001, "Timing for VLBI," presented at IVS TOW Meeting, Haystack, March 12, 2001. Available at: <http://www.gpstime.com/>

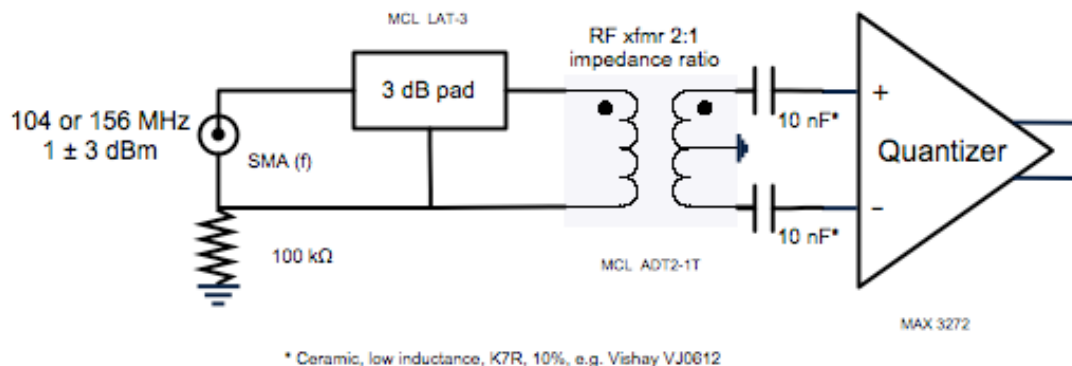
D'Addario, L. R., 2001, "ATA IF Processor: Requirements," Revision 2.1, 2001 July 14, Appendix A.

Harp, G. R., 2004, in "faster IP Processor control" thread of ATA Discussion Forum.

Thompson, A. R., Moran, J. M., and Swenson, G.W., Jr., 1986, *Interferometry and Synthesis in Radio Astronomy*, Wiley-Interscience.

Appendix A: Recommended RF Interfaces

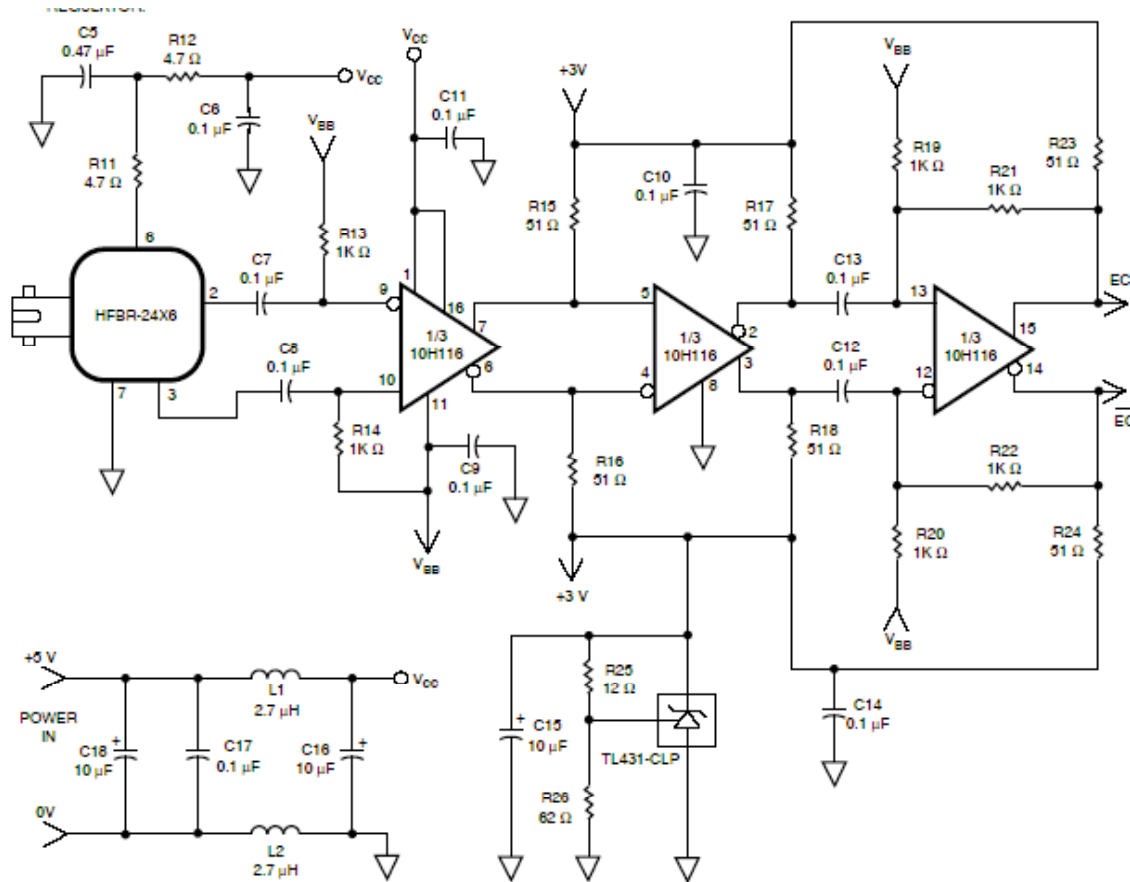
The 156.24 MHz and 104.16 MHz clock signals will be supplied as sinusoids on 50 Ω coaxial cable with male SMA connectors. The figure below shows the recommended RF interface to a high-speed, low-jitter quantizer that produces a CML digital clock suitable for distribution within the target rack-level system.



The input connector is on an isolated ground domain to prevent ground loops (at some cost in the shielding effectiveness of the coax). This isolated ground plane supports the 50 ohm transmission line to a 3 dB attenuator (MCL LAT-3, used to improve the input match) and thence to an RF transformer (MCL ADT2-1T). The 100 kohm resistor pulls the isolated ground to the local board ground whenever the input port is left open. The transformer converts the sinusoid from an unbalanced 50 ohm signal to a balanced 100 ohm signal and also provides a barrier to common mode noise. The center tap of the secondary is grounded to the local quantizer ground reference. The quantizer (limiting amp) requires AC coupling for proper operation (see manufacturer's datasheet); the coupling capacitors should be low-inductance K7R ceramic devices, *e.g.* Vishay VJ0612. The quantizer shown (MAX 3272) is the same device used in the multimode fiber links used for data transport in the Digital Signal Path. Driven at the recommended level (corresponding to -2 to $+4$ dBm at the input connector) the typical jitter of the resulting clock should be 3 ps rms. In order to maintain the low equivalent input noise of this quantizer (220 μ V rms) the power supply should carefully decoupled from any extraneous digital noise (noise on Vcc should be < 1 mV rms). For this application the offset correction capacitor can be omitted (pins CAZ1 and CAZ2 open).

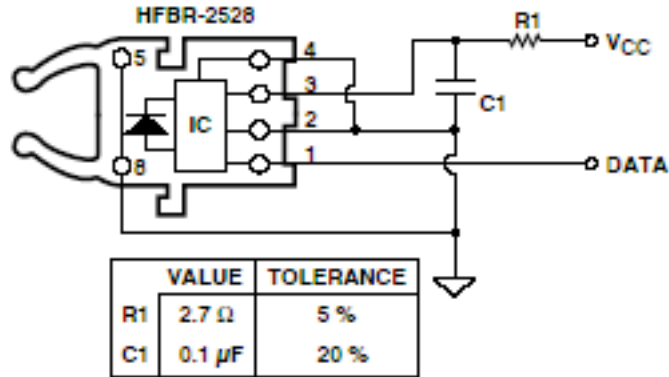
Appendix B: Recommended Optical Interfaces

The “fast synch” signal is a 406.875 kHz square wave with $50 \pm 1\%$ duty cycle distributed over 850 nm multimode fiber with ST connectors. The recommended optical-to-electrical interface device is the Agilent HFBR-2416TC, using one of the circuits from Agilent Application Note AB 78. An example for a positive ECL output is the following:



Please see AB78 for further details. Other connector options (SMA or FC) are available; please consult with the author if you plan to use one of these alternatives.

The ATA 1PPS is 1 Hz pulse with rise-time and jitter $\ll 1 \mu\text{s}$ and maintained to be within $\leq 10 \mu\text{s}$ of integral UTC seconds. It is distributed over 1 mm plastic optical fiber at 650 nm with Versatile Link connectors. These links are relatively low speed ($\sim 10 \text{ Mb}$) but allow DC coupled, arbitrary duty cycle TTL data to be used, which is desirable to keep the implementation of these low-duty-cycle pulses simple. The recommended optical-to-electrical interface device is the Agilent HFBR-2528, using the circuit from the Agilent HFBR-0508 series datasheet:



Capacitor C1 should be a low inductance ceramic chip type.